

Appendix E: BERT Search Hyperparameters

A Core Algorithm

These are the default high-level settings used by the encoder structured-pruning search.

Parameter	Value
Population size	120
Number of generations	80
Elite size	15
Number of islands	1

B Archives

The archives preserve high-quality candidates and prevent repeated macro signatures from dominating the stored final-ists.

Parameter	Value
Hall of Fame capacity	50
Hall of Fame max per macro signature	3
Pareto archive size	100

C Elites and Offspring

Elitism preserves the strongest candidates, while the offspring schedule shifts toward larger edits as stagnation increases.

Parameter	Value
Elite size	15
Per-island elites	15
Micro/macro split	0.35 initial; 0.80 maximum; cap reached at stagnation 15
Crossover rate	0.25 micro, 0.65 macro, 0.40 default

D Stagnation

Stagnation increases the probability of broader structural edits and can trigger additional full evaluations.

Parameter	Value
Restart threshold	stagnation ≥ 20
Extra full evals	24

E EXP3-IX Bandit

Each operator starts from a positive weight. The sampling probability mixes the learned distribution with uniform exploration.

Parameter	Value
Exploration coefficient γ	0.07
Initial weights	$w[o] = 1.0$ for all operators
Probability	$(1 - \gamma) w[o] / \sum_j w[j] + \gamma / K$
Update	$w[o] \leftarrow w[o] \exp\left(\gamma \frac{r}{p + \gamma} \frac{1}{K}\right)$
Overflow clip	normalise when $\max_o w[o] > 10^6$

F Island Operator Priors

Each island corresponds to a broad allocation style. The table below each label lists only the operators whose initial weights differ from the default value 1.0.

Head-heavy, low-neuron. This island favours moving capacity toward attention heads while keeping the feed-forward allocation small.

Operator	Weight
budget_neutral_swap_neurons_for_head	3.0
budget_neutral_swap_head	2.0
head_neuron_trade	1.5
layer_rewire	1.0
jitter	1.0
equilibrium_mutate	0.8

Head-heavy, mid-neuron. This island still favours attention, but keeps enough neuron capacity for mixed head–neuron refinements.

Operator	Weight
budget_neutral_swap_neurons_for_head	2.0
budget_neutral_swap_head	1.8
budget_neutral_swap_neurons	1.4
equilibrium_mutate	1.4
head_neuron_trade	1.2
jitter	1.0

Balanced. This island starts near an even attention and feed-forward allocation, so its priors favour local swaps and equilibrium-preserving moves.

Operator	Weight
budget_neutral_swap_head	1.8
budget_neutral_swap_neurons	1.8
equilibrium_mutate	1.6
group_reallocate	1.2
macro_budget_shift	1.2
jitter	1.0

Neuron-heavy. This island favours keeping more feed-forward capacity and uses operators that move budget from heads into neuron blocks.

Operator	Weight
budget_neutral_swap_head_for_neurons	3.0
budget_neutral_swap_neurons	2.2
neuron_rebalance	1.6
group_reallocate	1.2
jitter	1.0

Layer-sparse. This island explores architectures with fewer active layers and stronger depth-level edits.

Operator	Weight
drop_layer	3.0
drop_multi	2.5
layer_rewire	2.0
macro_budget_shift	1.8
budget_neutral_swap_head	1.2
budget_neutral_swap_neurons_for_head	1.2
jitter	1.0